

Ablation Results & Final Experiment

Luca Würker

August 2026

Protocol. Nasdaq-100 point-in-time panel, 2010–2026, survivorship-bias-free. Research uses data to mid-2021; 2021–26 is consumed as ten six-month blocks, each scored *before* the researcher may adapt to it — a five-year honest walk-forward record per run. All numbers are signal-level ICs; no position construction or costs.

Baseline. The full configuration: LLM ideation grounded in a knowledge graph of 1,723 papers (eight mechanism communities), candidates compiled and scored by a deterministic harness on four objectives (marginal value to the book, independence, parsimony, structural novelty), twenty generations of LLM mutation/crossover under progressive data reveal. 57 factors, ~900 scored candidates, \$250. Each arm below removes or adds one ingredient; everything else is held fixed.

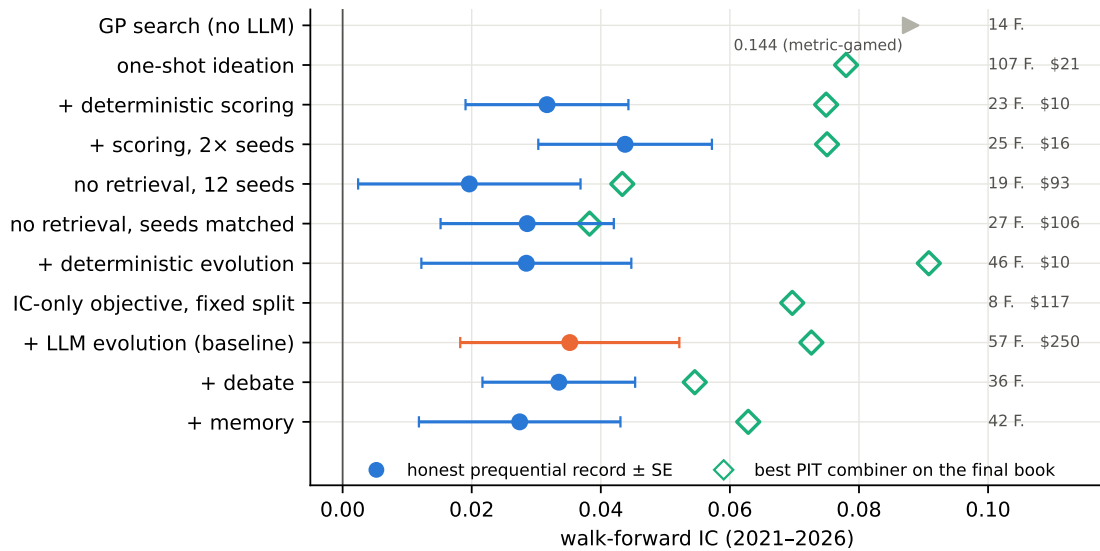


Figure 1: The ablation ladder. Dots: honest walk-forward record (mean of the ten block ICs $\pm$ SE). Diamonds: best point-in-time combiner applied to the arm’s final book. Right margin: book size and LLM cost.

Arm	Candidate evaluation	WF IC	PIT best	Book	Cost
GP (no LLM)	4-axis, incremental	0.144 *	—	14	\$0
One-shot	none — keep all	—	0.078	107	\$21
+ det. scoring	4-axis at admission	+0.032	0.075	23	\$10
+ scoring, 2× seeds	4-axis at admission	+0.044	0.075	25	\$16
No retrieval (96 seeds)	4-axis + reveal	+0.029	0.038	27	\$106
+ det. evolution	4-axis + reveal	+0.028	0.091	46	\$10
IC-only, fixed split	VAL IC only	—	0.070	8	\$117
Full LLM evolution	4-axis + reveal	+0.035	0.073	57	\$250
+ debate	4-axis (indep. axis inactive)	+0.033	0.055	36	—
+ memory	4-axis (indep. axis inactive)	+0.027	0.063	42	\$206

Table 1: * metric-gamed (integrated price/fundamental levels), not alpha. Block SE $\approx$ 0.012–0.019. “One-shot” keeps every factor that compiles; the scoring arms evaluate each seed on the four objectives sequentially at admission (against the book built so far) and then re-score, prune and curate for twenty generations without generating children.

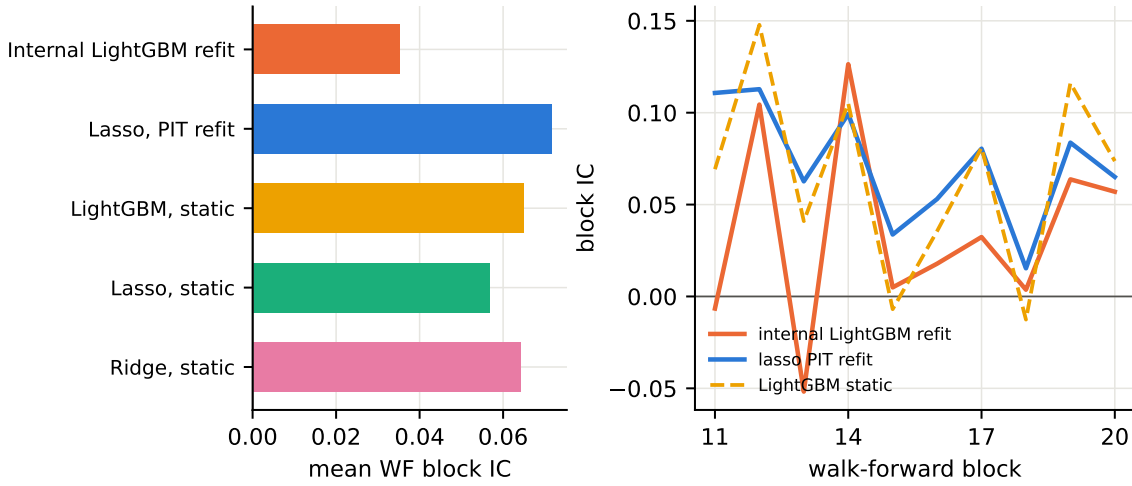


Figure 2: Baseline book, five ways to combine it. The prequential LightGBM refit the search optimises against (orange) is the weakest deployment combiner; a point-in-time lasso refit doubles it and is positive in every block. The same ordering holds in every arm.

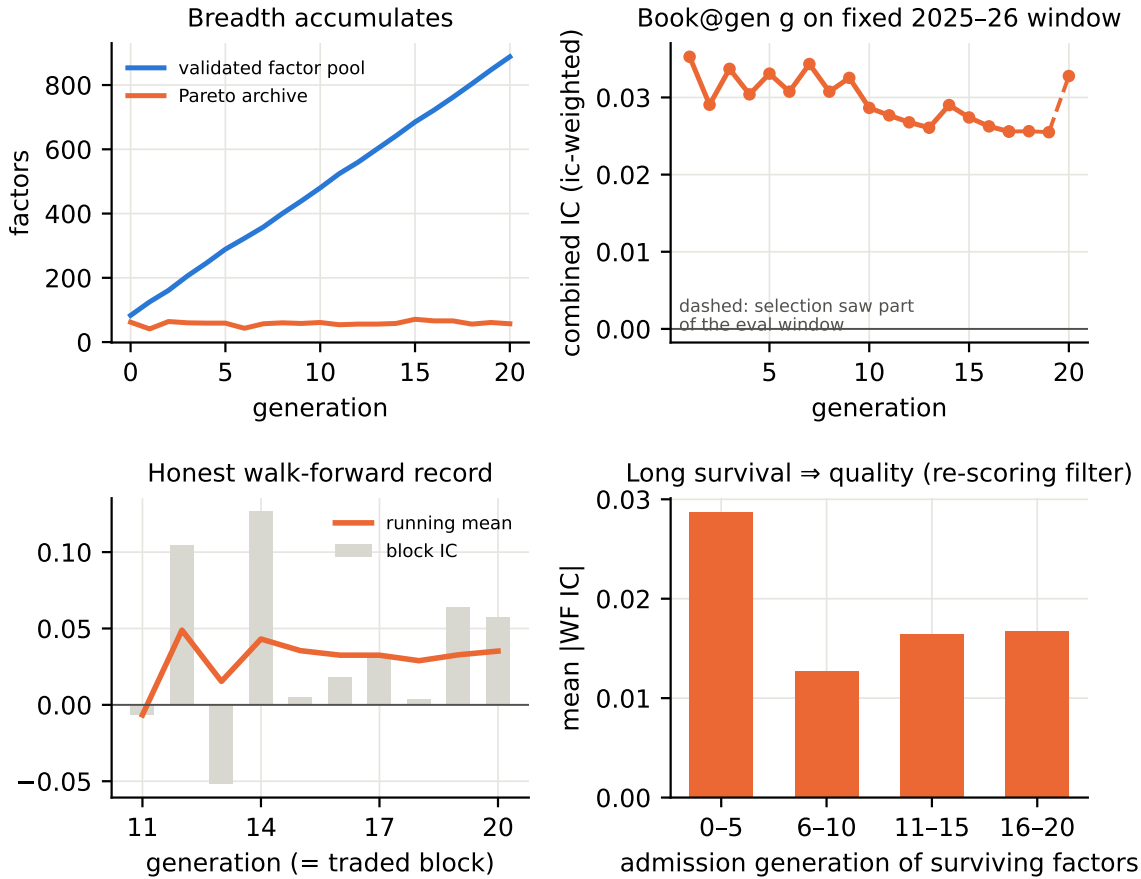


Figure 3: What actually changes over the evolution (baseline). Breadth accumulates (pool 83→890) while the archive holds at its cap; the archive as of each generation, recombined identically on a fixed 2025–26 window, is flat — the generation-1 book was already as good; the honest record stabilises at $\approx +0.035$; and the earliest admits — re-scored up to twenty times on growing, partly unseen windows — carry twice the realised $|IC|$ of later cohorts.

Takeaways

1. **Per-factor edge transfers out of sample.** Fit-window versus walk-forward per-factor IC: retention $r = 0.94$, only 18% of baseline factors flip sign, mean $|IC|$ unchanged ($0.019 \rightarrow 0.018$). The *combiner* is the fragile part: point-in-time linear recombination doubles the internal boosted-tree score (0.073 vs. 0.035) — so results should be reported combiner-robust, not on one combiner’s IC.
2. **Ideation + deterministic scoring carries the record.** Seeding once and letting the harness score, prune and curate — no evolution — matches the \$250 full-evolution record at \$10–16; doubling the seed budget gives the family’s best honest record ($+0.044$). In-loop LLM mutation, debate and cross-run memory add nothing measurable.
3. **Every naive objective gets gamed.** The prior-free GP maximises its reward with integrated price/fundamental levels and, after a targeted stationarity gate, evades it with near-boundary proxies (Goodhart). Scoring by validation IC alone on a fixed split shows textbook winner’s curse: promised median $|IC|$ $0.040 \rightarrow 0.025$ delivered, five of eight survivors sign-unstable, the archive collapsed to one factor per group. Multi-objective selection + progressive reveal are what make individual factors trustworthy.
4. **Retrieval grounding doubles deployable value, not the headline.** With seeding matched, the no-retrieval arm ties the baseline’s record ($+0.029$ vs. $+0.035$) but produces the family’s weakest factors (median $|IC|$ 0.005 , 44% sign flips, lowest effective dimension) and recombines to 0.038 versus

0.073. (The first no-retrieval runs were confounded — underseeded by a since-fixed timeout bug; the control run resolves it.)

5. **Evolution buys breadth and sign stability, not mean IC.** The generation-1 book already scores as well on a fixed future window as any later one; survival through repeated re-scoring is the effective quality filter. Deterministic jitter evolution captures this cheaply: the best point-in-time book of the family (0.091) at \$10.
6. **All results are single-seed; differences within ± 0.01 are noise.** This motivates the design of the final experiment.

Final experiment: LLM comparison

Last remaining study: several research LLMs on one fixed configuration. Chosen configuration: **grounded ideation + deterministic scoring** — 192 graph-grounded seed ideas (24 per mechanism community), no evolutionary children, the full walk-forward scoring machinery; the knowledge graph is frozen so every model sees identical context; arms are seed-matched, with realised cost reported as a secondary result.

- **It isolates what varies between models** — idea quality and code quality — judged by a fixed deterministic harness. Full evolution would spend most of each budget on machinery the ablations show adds nothing, diluting the model contrast with selection noise.
- **It is the family's efficient frontier:** best honest record (+0.044) and point-in-time 0.075, on par with the \$250 baseline.
- **Statistical power per dollar:** ~\$20 per run allows two to three seeds per model — roughly \$250 for six models, the price of one full-evolution run — addressing the single-seed weakness directly.
- **Everything downstream exists:** reference runs on the current model, and the standardised evaluation (prequential record, combiner race, per-factor |IC|, sign stability, effective dimension) applies unchanged.

Planned comparisons: the current reference model plus frontier-class and mid-tier alternatives from the major providers, two seeds each; identical prompts, graph snapshot, budget cap per arm. Metrics: honest walk-forward record, best-combiner book IC, factor validity rate, per-factor |IC| and sign stability, diversity, and cost per validated factor. Before launch: fix the coverage bug that silently disabled the independence objective in the debate arms.